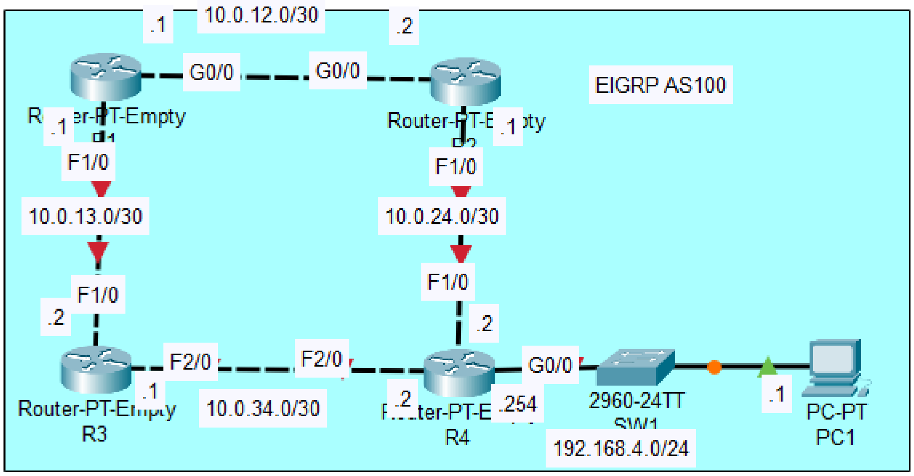

Topology

1. Configure the appropriate hostnames and IP addresses on each device. Enable router interfaces.

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int g0/0
R1(config-if)#ip address 10.0.12.1 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

R1(config-if)#int f1/0
R1(config-if)#ip address 10.0.13.1 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int f1/0
R2(config-if)#ip address 10.0.24.1 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

R2(config-if)#int g0/0
R2(config-if)#ip address 10.0.12.2 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#int f1/0
R3(config-if)#ip address 10.0.13.2 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R3(config-if)#int f2/0
R3(config-if)#ip address 10.0.34.1 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet2/0, changed state to up
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R4
R4(config)#int f1/0
R4(config-if)#ip address 10.0.24.1 255.255.255.252
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
%IP-4-DUPADDR: Duplicate address 10.0.24.1 on FastEthernet1/0, sourced by 00E0.8F32.9DC3
%IP-4-DUPADDR: Duplicate address 10.0.24.1 on FastEthernet1/0, sourced by 00E0.8F32.9DC3

R4(config-if)#int f2/0
R4(config-if)#ip address 10.0.34.2 255.255.255.252
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R4(config-if)#int g0/0
R4(config-if)#ip address 192.168.4.254 255.255.255.0
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

2. Configure a loopback interface on each router (1.1.1.1/32 for R1, 2.2.2.2/32 for R2, etc.)

```
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#do sh ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.12.1       YES manual up                    up
FastEthernet1/0        10.0.13.1       YES manual up                    up
FastEthernet2/0        unassigned      YES unset  administratively down down
Loopback0              1.1.1.1         YES manual up                    up
```

```
R2(config)#interface loopback 0

R2(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R2(config-if)#ip address 2.2.2.2 255.255.255.255
R3(config)#interface loopback 0

R3(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R3(config-if)#ip address 3.3.3.3 255.255.255.255
R4(config)#int loopback 0

R4(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R4(config-if)#ip address 4.4.4.4 255.255.255.255
```

3. Configure EIGRP on each router. Disable auto-summary. Enable EIGRP on each interface (including loopback interfaces). Configure passive interfaces where appropriate (including loopback interfaces).

```
R1(config)#router eigrp 100
R1(config-router)#network 0.0.0.0 255.255.255.255
R1(config-router)#no auto-summary
R1(config-router)#passive-interface l0
R2(config)#router eigrp 100
R2(config-router)#networ 0.0.0.0 255.255.255.255
R2(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.12.1 (GigabitEthernet0/0) is up: new adjacency

R2(config-router)#no auto-summary
R2(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.12.1 (GigabitEthernet0/0) resync: summary configured

R2(config-router)#passive-interface l0
R3(config)#router eigrp
% Incomplete command.
R3(config)#router eigrp 100
R3(config-router)#network 0.0.0.0 255.255.255.255
R3(config-router)#no auto-summary
R3(config-router)#[assove-interface l0
                  ^
% Invalid input detected at '^' marker.

R3(config-router)#passive-interface l0
R4(config)#router eigrp 100
R4(config-router)#network 0.0.0.0 255.255.255.255
R4(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.34.1 (FastEthernet2/0) is up: new adjacency

R4(config-router)#no auto-summary
R4(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.34.1 (FastEthernet2/0) resync: summary configured

R4(config-router)#passive-interface l0
R4(config-router)#passive interface g0/0
                          ^
% Invalid input detected at '^' marker.

R4(config-router)#passive-interface g0/0
```

4. Configure R1 to perform unequal-cost load-balancing when sending network traffic to 192.168.4.0/24

```
R1(config)#router eigrp 100
R1(config-router)#variance 2
R1(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.12.2 (GigabitEthernet0/0) is up: new adjacency
```